

Week – 1 : MCQs on Python Packages for ML

Questions:

1. **Machine learning is a subset of artificial intelligence that focuses on enabling machines to learn and make decisions without being explicitly programmed.**
 - a) True
 - b) False

Correct Answer: a) True

Explanation: Machine learning is indeed a subset of AI that allows systems to learn from data and make decisions or predictions without explicitly following hard-coded rules.

2. **Which type of machine learning is primarily used for clustering analysis?**
 - a) Supervised Learning
 - b) Unsupervised Learning
 - c) Reinforcement Learning
 - d) Semi-supervised Learning

Correct Answer: b) Unsupervised Learning

Explanation: Clustering is a common task in unsupervised learning where the goal is to group data points into clusters without pre-defined labels.

3. **Which of the following is NOT a challenge in machine learning?**
 - a) Handling missing data
 - b) Ensuring high computational resources
 - c) Creating handcrafted rules for each task
 - d) Managing model overfitting

Correct Answer: c) Creating handcrafted rules for each task

Explanation: Machine learning avoids the need to create handcrafted rules, as the model learns patterns from data. The other options are genuine challenges in machine learning.

4. **Which of the following situations indicates that a model is underfitting the data?**
 - A) The model has high training accuracy but low testing accuracy.
 - B) The model performs poorly on both training and testing data.
 - C) The model achieves high testing accuracy but low training accuracy.
 - D) The model captures too much noise in the training data.

Correct Answer: B

Explanation:

Underfitting happens when the model is too simplistic and unable to capture the

underlying patterns in the data, leading to poor performance on both the training and testing datasets.

5. What happens when a model has **high bias** in the context of the bias-variance tradeoff?
- A) The model captures noise in the training data, leading to overfitting.
 - B) The model is too simple and performs poorly on the training data.
 - C) The model has low error on the test data and generalizes well.
 - D) The model has high variance and performs inconsistently on new data.

Correct Answer: B

Explanation:

High bias indicates that the model is too simple (e.g., a linear model for non-linear data), leading to underfitting and poor performance on both training and testing datasets. This is due to the inability to capture the underlying complexities of the data.

6. Which programming languages are supported by Jupyter Notebook?
- a) Python, R, Julia
 - b) C++, Java, PHP
 - c) Ruby, Perl, Swift
 - d) HTML, CSS, JavaScript

Correct Answer: a) Python, R, Julia

7. Which file format does Jupyter Notebook use to store the notebooks?
- a) .py
 - b) .ipynb
 - c) .jnb
 - d) .pynb

Correct Answer: b). ipynb

8. What is the default data type of the value returned by the input () function in Python?
- a) Integer
 - b) Float
 - c) String
 - d) Boolean

Correct Answer: c) String

9. What will be the output of the following Python function if the random module has already been imported?

random.randint (3,7)

- a) Any integer between 3 and 7, including 7
- b) Any integer between 3 and 7, excluding 7
- c) Any inter closest to 3 and 7
- d) Error

Correct Answer: a) Any integer between 3 and 7, including 7

10. What is the command used to read a csv file in python?

- a) pd.Read_Csv ("file path")
- b) pd.Read_csv ("file path")
- c) pd.READ_CSV ("file path")
- d) pd.read_csv ("file path")

Correct Answer: d) pd.read_csv ("file path")